SOLUTIONS: SYSTEMS IDENTIFICATION AND LEARNING

1. a) Using the one-step forward shift operator z, we get:

$$\begin{cases} (z-0.3)x_1(t) = 0.5x_2(t) + zv(t) \\ (z+0.1)x_2(t) = 0.2x_1(t) \\ y(t) = x_1(t) \end{cases}$$

Eliminating the variable x_2 from equations above, after some algebra, one gets:

$$(1-0.2z^{-1}-0.13z^{-2})y(t) = (1+0.1z^{-1})v(t)$$

which can be equivalently re-written in canonical form as:

$$y(t) = \frac{z^2+0.1z}{z^2-0.2z-0.13}v(t)$$

The poles p_1, p_2 of the polynomial $z^2-0.2z-0.13$ are $p_1 \simeq -0.27$ and $p_2 \simeq 0.47$. Since they are both located strictly inside the unit circle, we conclude that the stochastic process $y(\cdot)$ is stationary.

Therefore, the stationary ARMA model takes on the form:

$$y(t) = 0.2y(t-1) + 0.13y(t-2) + v(t) + 0.1v(t-1).$$

[4 Marks]

b) Based on the ARMA model determined in the answer to Question a) and owing to the fact that the stochastic process $y(\cdot)$ is stationary and that the ARMA model is in canonical form, we proceed as follows.

We write

$$A(z)y(t) = C(z)v(t)$$

where

$$A(z) = 1-0.2z^{-1}-0.13z^{-2}; C(z) = 1+0.1z^{-1}.$$

Then, the optimal one-step ahead prediction takes on the form

$$\widehat{y}(t+1|t) = \frac{C(z)-A(z)}{C(z)}y(t+1) = \frac{0.3z^{-1}+0.13z^{-2}}{1+0.1z^{-1}}y(t+1)$$

which can be rewritten as

$$\widehat{y}(t+1|t) = \frac{0.3+0.13z^{-2}}{1+0.1z^{-1}}y(t)$$

thus yielding the expression of the optimal one-step ahead predictor:

$$\widehat{y}(t+1|t) = -0.1\widehat{y}(t|t-1) + 0.3y(t) + 0.13y(t-1)$$

[4 Marks]

Answers

c) The prediction error $\varepsilon_1(t)$ associated with the optimal one-step ahead predictor determined in the answer to Question b) is given by

$$\varepsilon_1(t) = y(t+1) - \hat{y}(t+1|t) = v(t+1)$$

Hence, the variance can be trivially obtained as

$$\text{var}[\varepsilon_1(t)] = 1\,.$$

[3 Marks]

d) By carrying out two steps of the division of $C(z)$ by $A(z)$, the first step gives

$$\frac{C(z)}{A(z)} = \frac{1+0.1z^{-1}}{1-0.2z^{-1}-0.13z^{-2}} = 1 + z^{-1}\frac{0.3+0.13z^{-1}}{1-0.2z^{-1}-0.13z^{-2}}$$

and then, in the next step one gets

$$\frac{C(z)}{A(z)} = 1 + 0.3z^{-1} + z^{-2}\frac{0.19+0.039z^{-1}}{1-0.2z^{-1}-0.13z^{-2}}$$

Therefore, since $v(t) = A(z)/C(z)y(t)$, the optimal two-steps ahead predictor can be written as follows:

$$\widehat{y}(t+2|t) = \frac{0.19+0.039z^{-1}}{1+0.1z^{-1}}y(t)$$

which finally yields:

$$\widehat{y}(t+2|t) = -0.1\widehat{y}(t+1|t) + 0.19y(t) + 0.039y(t-1).$$

[3 Marks]

e) Owing to the answer to Question c), the two-step-ahead prediction error is given by

$$\varepsilon_2(t) = y(t+2) - \widehat{y}(t+2|t) = (1+0.3z^{-1})v(t+2)$$

Hence:

$$\varepsilon_2(t) = y(t+2) - \widehat{y}(t+2|t) = v(t+2) + 0.3v(t+1)$$

Thus, the variance can be computed as follows:

$$\text{var}[\varepsilon_2(t)] = \text{var}[v(t+2)] + 0.3^2\text{var}[v(t+1)] = 1 + 0.09 = 1.09\,.$$

[3 Marks]

f) The comparison between $\text{var}[\varepsilon_1(t)]$ and $\text{var}[\varepsilon_2(t)]$ gives

$$\text{var}[\varepsilon_2(t)] = 1.09 > 1 = \text{var}[\varepsilon_1(t)]$$

Accordingly, the variance of the error when the optimal one-step ahead predictor is used is smaller than when the optimal two-steps ahead predictor is used. This should be expected due to the fact that in the one-step ahead predictor, $\widehat{y}(t+1|t)$ is a function of samples closer in time to the predicted quantity than those used by the two-steps ahead predictor.

[3 Marks]

Answers

2. a) The process $y(\cdot)$ can be written as the sum of two uncorrelated processes $y_1(\cdot)$ and $y_2(\cdot)$, that is:

$$y(t) = y_1(t) + y_2(t) \text{ where } y_1(t) = \frac{8+18z^{-1}+4z^{-2}}{4+2z^{-1}}\omega(t);\ y_2(t) = \frac{2+5z^{-1}}{4+z^{-1}}\eta(t)$$

Therefore $y(\cdot)$ is stationary if the processes $y_1(\cdot)$ and $y_2(\cdot)$ are stationary. Both processes are described as a dynamical system fed with a white process. Hence, $y_1(\cdot)$ and $y_2(\cdot)$ are stationary if the respective dynamical models are asymptotically stable.

Hence, we write:

$$y_1(t) = \frac{8+18z^{-1}+4z^{-2}}{4+2z^{-1}}\omega(t) = \frac{8z^2+18z+4}{4z^2+2z}\omega(t)$$

The poles are the roots of the denominator $4z^2+2z$, that is:

$$p_1 = 0\ ; p_2 = -\frac{1}{2}$$

that are both located strictly inside the unit circle. Therefore, $y_1(\cdot)$ is a stationary process.

Moreover:

$$y_2(t) = \frac{2+5z^{-1}}{4+z^{-1}}\eta(t) = \frac{2z+5}{4z+1}\eta(t)$$

The pole is the root of the denominator $4z+1$, that is:

$$p_1 = -\frac{1}{4}$$

that is located strictly inside the unit circle. Therefore, also $y_2(\cdot)$ is a stationary process. In conclusion, the stochastic process $y(\cdot)$ is stationary.

To compute the variance $\text{var}[y(t)]$ of the process $y(\cdot)$ we first observe that, due to the uncorrelation of the stationary stochastic processes $y_1(\cdot)$ and $y_2(\cdot)$, we get:

$$\text{var}[y(t)] = \text{var}[y_1(t)] + \text{var}[y_2(t)] = \mathbb{E}[y_1(t)^2] + \mathbb{E}[y_2(t)^2]$$

Thus, we proceed by determining $\text{var}[y_1(t)]$ and $\text{var}[y_2(t)]$ separately.

Concerning $\text{var}[y_1(t)]$, we have:

$$y_1(t) = \frac{8(z+1/4)(z+2)}{4z(z+1/2)}\omega(t) = 2\frac{(z+\frac{1}{4})}{z}\cdot 2\cdot \underbrace{\frac{1}{2}\cdot\frac{z+2}{z+\frac{1}{2}}}_{\text{all-pass filter}}\omega(t)$$

$$= 4\frac{z+\frac{1}{4}}{z}\omega(t) = (4+z^{-1})\omega(t)$$

Therefore, y_1 is an MA process:

$$y_1(t) = (4+z^{-1})\omega(t) = 4\omega(t) + \omega(t-1).$$

Hence:

$$\text{var}[y_1(t)] = \frac{1}{5}(4^2+1) = \frac{17}{5} = 3.4$$

Concerning $\text{var}[y_2(t)]$, we have

$$y_2(t) = \frac{2+5z^{-1}}{4+z^{-1}}\eta(t)$$

Answers

$$\implies y_2(t)(4+z^{-1}) = (2+5z^{-1})\eta(t)$$
$$\implies 4y_2(t) + y_2(t-1) = 2\eta(t) + 5\eta(t-1)$$
$$\implies y_2(t) = -\frac{1}{4}y_2(t-1) + \frac{1}{2}\eta(t) + \frac{5}{4}\eta(t-1)$$

Thus:

$$\begin{aligned}
\text{var}[y_2(t)] &= \mathbb{E}[y_2(t)^2] \\
&= \mathbb{E}\left[\left(-\frac{1}{4}y_2(t-1) + \frac{1}{2}\eta(t) + \frac{5}{4}\eta(t-1)\right)^2\right] \\
&= \frac{1}{16}\underbrace{\mathbb{E}[y_2(t-1)^2]}_{V_{y_2}} + \frac{1}{4}\underbrace{\mathbb{E}[\eta(t)^2]}_{3} + \frac{25}{16}\underbrace{\mathbb{E}[\eta(t-1)^2]}_{3} - \frac{1}{4}\underbrace{\mathbb{E}[y_2(t-1)\eta(t)]}_{0} + \\
&\quad - \frac{5}{8}\underbrace{\mathbb{E}[y_2(t-1)\eta(t-1)]}_{\mathbb{E}[y_2(t)\eta(t)]} + \frac{5}{4}\underbrace{\mathbb{E}[\eta(t)\eta(t-1)]}_{0} \\
&= \frac{1}{16}\text{var}[y_2(t)] + \frac{3}{4} + \frac{75}{16} - \frac{5}{8}\mathbb{E}[y_2(t)\eta(t)]
\end{aligned}$$

Moreover,

$$\begin{aligned}
\mathbb{E}[y_2(t)\eta(t)] &= \mathbb{E}\left[\left(-\frac{1}{4}y_2(t-1) + \frac{1}{2}\eta(t) + \frac{5}{4}\eta(t-1)\right)\eta(t)\right] \\
&= \frac{1}{2}\underbrace{\mathbb{E}[\eta(t)^2]}_{3} = \frac{3}{2}
\end{aligned}$$

Therefore,

$$\text{var}[y_2(t)] = \frac{1}{16}V_{y_2} + \frac{3}{4} + \frac{75}{16} - \frac{5}{8}\cdot\frac{3}{2} = \frac{24}{5} = 4.8$$

Finally,

$$\text{var}[y(t)] = \text{var}[y_1(t)] + \text{var}[y_2(t)] = \frac{17}{5} + \frac{24}{5} = \frac{41}{5} = 8.2$$

[7 Marks]

b) i) Since the correlation function $\gamma_{\widetilde{\omega}}(\tau)$ is an even function, we focus on $\tau \geq 0$. Hence:

$$\begin{aligned}
\gamma_{\widetilde{\omega}}(\tau) &= \mathbb{E}[\widetilde{\omega}(t)\widetilde{\omega}(t-\tau)] \\
&= \mathbb{E}\left[\left(\sum_{k=0}^{\infty}\left(\frac{1}{3}\right)^k e(t-k)\right)\left(\sum_{k=0}^{\infty}\left(\frac{1}{3}\right)^k e(t-k-\tau)\right)\right]
\end{aligned}$$

As $e(\cdot)$ is a white process, we get:

$$\begin{aligned}
\gamma_{\widetilde{\omega}}(\tau) &= \mathbb{E}\left[\sum_{k=0}^{\infty}\left(\frac{1}{3}\right)^k\left(\frac{1}{3}\right)^{k+\tau} e(t-k)^2\right] \\
&= \sum_{k=0}^{\infty}\left(\frac{1}{3}\right)^k\left(\frac{1}{3}\right)^{k+\tau}\underbrace{\mathbb{E}[e(t-k)^2]}_{7/10} \\
&= \frac{7}{10}\left(\frac{1}{3}\right)^{\tau}\sum_{k=0}^{\infty}\left(\frac{1}{3}\right)^{2k}
\end{aligned}$$

Answers

$$= \frac{7}{10}\left(\frac{1}{3}\right)^{\tau}\frac{1}{1-\frac{1}{9}}$$
$$= \frac{7}{10}\left(\frac{1}{3}\right)^{\tau}\frac{9}{8}$$
$$= \frac{7\cdot 3^{2-\tau}}{80}$$

Finally, we can easily extend this result to the negative values of τ thus obtaning:

$$\gamma_{\widetilde{\omega}}(\tau) = \frac{7\cdot 3^{2-|\tau|}}{80}$$

[7 Marks]

ii) By following the same approach followed in answering Question 2-a), we note that the process $\widetilde{y}(\cdot)$ can be written as the sum of two independent processes $\widetilde{y}_1(\cdot)$ and $y_2(\cdot)$:

$$\widetilde{y} = \widetilde{y}_1(t) + y_2(t)$$

with

$$\widetilde{y}_1(t) = \frac{8+18z^{-1}+4z^{-2}}{4+2z^{-1}}\widetilde{\omega}(t)$$

and $y_2(\cdot)$ being the same obtaoned in the answer to Question 2-a). Hence, we just focus on $\widetilde{y}_1(\cdot)$. Clearly, $\widetilde{y}_1(\cdot)$ and $y_2(\cdot)$ are still stationary, uncorrelated, and zero-mean. Therefore,

$$\text{var}[\widetilde{y}(t)] = \text{var}[\widetilde{y}_1(t)] + \text{var}[y_2(t)]$$

We compute $\text{var}[\widetilde{y}_1(t)]$ as follows:

$$\begin{aligned}\text{var}[\widetilde{y}_1(t)] &= \mathbb{E}\left[(4\widetilde{\omega}(t)+\widetilde{\omega}(t-1))^2\right]\\ &= \mathbb{E}\left[16\widetilde{\omega}(t)^2+\widetilde{\omega}(t-1)^2+8\widetilde{\omega}(t)\widetilde{\omega}(t-1)\right]\\ &= 16\gamma_{\widetilde{\omega}}(0)+\gamma_{\widetilde{\omega}}(0)+8\gamma_{\widetilde{\omega}}(1)\end{aligned}$$

Recall that $\gamma_{\widetilde{\omega}}(0)$ and $\gamma_{\widetilde{\omega}}(1)$ denote the first two samples of the correlation function $\gamma_{\widetilde{\omega}}(\tau)$ of $\widetilde{\omega}(\cdot)$ computed in the answer to Question 2b)-i). Therefore:

$$\text{var}[\widetilde{y}_1(t)] = 17\cdot\frac{7\cdot 3^2}{80}+8\cdot\frac{7\cdot 3^1}{80} = \frac{1239}{80} = 15.4875$$

thus yielding (recall from the answer to Question 2-a) that $\text{var}[y_2(t)] = 24/5$):

$$\text{var}[\widetilde{y}(t)] = \text{var}[\widetilde{y}_1(t)] + \text{var}[y_2(t)] = \frac{1239}{80}+\frac{24}{5} = \frac{1303}{80} \simeq 17.29$$

[6 Marks]

Answers

3. a) We consider the difference equation describing the stochastic model $\mathscr{M}_1(b_1)$:

$$v(t) = b_1 v(t-1) + \varepsilon_1(t)$$

For $t = 2,3,\ldots,N$, we introduce the residual process given by the prediction error

$$r_1(t) = v(t) - b_1 v(t-1)\,.$$

The best estimate (in the least-squares sense) $\widehat{b}_1$ of the unknown true parameter b_1 can be obtained by minimising the sum of squared residuals, that is:

$$\widehat{b}_1 = \frac{\sum_{t=2}^{N}[v(t)v(t-1)]}{\sum_{t=2}^{N}[v(t-1)]^2}$$

Using the estimate $\widehat{b}_1$ of b_1, an estimate $\widehat{\sigma}_1^2$ of the variance σ_1^2 is given by:

$$\widehat{\sigma}_1^2 = \frac{1}{N-1}\sum_{t=2}^{N}[v(t) - \widehat{b}_1 v(t-1)]^2\,.$$

[6 Marks]

b) Now, we consider the difference equation describing the stochastic model $\mathscr{M}_2(a_1,a_2)$:

$$v(t) = a_1 v(t-1) + a_2 v(t-2) + \varepsilon_2(t),\ \varepsilon_2(t) \sim WN(0,\sigma_2^2)$$

For $t = 3,4,\ldots,N$, we introduce the residual process given by the prediction error

$$r_2(t) = v(t) - a_1 v(t-1) - a_2 v(t-2)\,.$$

The best estimates (in the least-squares sense) $\widehat{a}_1,\widehat{a}_2$ of the unknown true parameters a_1,a_2 can be obtained by minimising the sum of squared residuals, that is:

$$\begin{bmatrix}\widehat{a}_1\\ \widehat{a}_2\end{bmatrix} = \left[\sum_{t=3}^{N}\begin{bmatrix}v(t-1)\\ v(t-2)\end{bmatrix}\begin{bmatrix}v(t-1) & v(t-2)\end{bmatrix}\right]^{-1}\sum_{t=3}^{N}\begin{bmatrix}v(t-1)\\ v(t-2)\end{bmatrix}v(t).$$

Using the estimates $\widehat{a}_1,\widehat{a}_2$ of a_1,a_2, an estimate $\widehat{\sigma}_2^2$ of the variance σ_2^2 is given by:

$$\widehat{\sigma}_2^2 = \frac{1}{N-2}\sum_{t=3}^{N}[v(t) - \widehat{a}_1 v(t-1) - \widehat{a}_2 v(t-2)]^2.$$

[7 Marks]

c) Finally, we consider the difference equation describing the stochastic model $\mathscr{M}_3(c_1)$:

$$v(t) = c_1 u(t-1) + \eta(t)$$

The difference between stochastic model $\mathscr{M}_3(c_1)$ and models $\mathscr{M}_1(b_1)$ and $\mathscr{M}_2(a_1,a_2)$ considered in the answers to Questions a) and b), respectively, is the absence of an "auto-regressive" term. Since the least-squares approach just minimises the sum of the squares of properly defined prediction errors, the procedure to be carried out is the same as in the answer to Question a).

Answers

Thus, for $t = 2, 3, \ldots, N$, we introduce the residual process given by the prediction error

$$r_3(t) = v(t) - c_1 v(t-1)\,.$$

The best estimate (in the least-squares sense) $\widehat{c}_1$ of the unknown true parameter c_1 can be obtained by minimising the sum of squared residuals, that is:

$$\widehat{c}_1 = \frac{\sum_{t=2}^{N} [v(t)u(t-1)]}{\sum_{t=2}^{N} [u(t-1)]^2}$$

Using the estimate $\widehat{c}_1$ of c_1, an estimate $\widehat{\sigma}_3^2$ of the variance σ_3^2 is given by:

$$\widehat{\sigma}_3^2 = \frac{1}{N-1} \sum_{t=2}^{N} [v(t) - \widehat{c}_1 v(t-1)]^2\,.$$

[7 Marks]

Answers

4. a) Consider a linear estimator for T of the form stated in the text of Question 4-a):

$$\widehat{T}(x_1,x_2,x_3) = \alpha x_1 + \beta x_2 + \gamma x_3 + \delta,$$

where $\alpha,\beta,\gamma,\delta \in \mathbb{R}$ are parameters to be determined. The application of the expected value operator to both sides gives:

$$\begin{aligned}\mathbb{E}[\widehat{T}(x_1,x_{2,3})] &= \alpha\mathbb{E}(x_1) + \beta\mathbb{E}(x_2) + \gamma\mathbb{E}(x_3) + \delta \\ &= (\alpha+\beta+\gamma)T + \alpha b_1 + \beta b_2 + \gamma b_3 + \delta\,.\end{aligned}$$

Then, the estimator $\widehat{T}(x_1,x_2,x_3)$ is unbiased if

$$\alpha+\beta+\gamma = 1$$

and

$$\alpha b_1 + \beta b_2 + \gamma b_3 + \delta = 0\,.$$

[3 Marks]

b) We compute the variance $\mathrm{var}[\widehat{T}]$ of the estimator as follows:

$$\begin{aligned}\mathrm{var}[\widehat{T}(x_1,x_2,x_3)] &= \mathbb{E}\left\{[\alpha(x_1-T-b_1)+\beta(x_2-T-b_2)+\gamma(x_3-T-b_3)]^2\right\} \\ &= \alpha^2\mathrm{var}(x_1) + \beta^2\mathrm{var}(x_2) + \gamma^2\mathrm{var}(x_3) \\ &+ 2\alpha\beta\mathbb{E}[(x_1-T-b_1)(x_2-T-b_2)] \\ &+ 2\alpha\gamma\mathbb{E}[(x_1-T-b_1)(x_3-T-b_3)] \\ &+ 2\beta\gamma\mathbb{E}[(x_2-T-b_2)(x_3-T-b_3)]\end{aligned}$$

Since the measurements x_1, x_2, and x_3 are mutually uncorrelated, it follows that:

$$\begin{aligned}\mathbb{E}[(x_1-T-b_1)(x_2-T-b_2)] &= 0 \\ \mathbb{E}[(x_1-T-b_1)(x_3-T-b_3)] &= 0 \\ \mathbb{E}[(x_1-T-b_1)(x_3-T-b_3)] &= 0\end{aligned}$$

Then:

$$\mathrm{var}[\widehat{T}(x_1,x_2,x_3)] = \alpha^2\mathrm{var}(x_1) + \beta^2\mathrm{var}(x_2) + \gamma^2\mathrm{var}(x_3)$$

Owing to the fact that:

$$x_1 \sim \mathcal{N}(T+b_1,\sigma_1^2)\,,\quad x_2 \sim \mathcal{N}(T+b_2,\sigma_2^2)\,,\quad x_3 \sim \mathcal{N}(T+b_3,\sigma_3^2)\,,$$

we finally obtain the following expression for $\mathrm{var}[\widehat{T}]$:

$$\mathrm{var}[\widehat{T}(x_1,x_2,x_3)] = \alpha^2\sigma_1^2 + \beta^2\sigma_2^2 + \gamma^2\sigma_3^2\,.$$

[4 Marks]

c) To ensure unbiasedness of the estimator, the following constraints hold according to the answer to Question a):

$$\alpha+\beta+\gamma = 1$$

and

$$\alpha b_1 + \beta b_2 + \gamma b_3 + \delta = 0\,.$$

Answers

Therefore, it follows that:

$$\begin{aligned}\text{var}(\widehat{T}) &= \alpha^2\sigma_1^2 + \beta^2\sigma_2^2 + (1-\alpha-\beta)^2\sigma_3^2 \\ &= (\sigma_1^2+\sigma_3^2)\alpha^2 + (\sigma_2^2+\sigma_3^2)\beta^2 + \sigma_3^2 - 2\sigma_3^2\alpha - 2\sigma_3^2\beta + 2\sigma_3^2\alpha\beta\end{aligned}$$

To minimise $\text{var}(\widehat{T})$ with respect to α, β, one first computes:

$$\begin{aligned}\frac{\partial}{\partial\alpha}\text{var}(\widehat{T}) &= (\sigma_1^2+\sigma_3^2)\alpha + \sigma_3^2\beta - \sigma_3^2 \\ \frac{\partial}{\partial\beta}\text{var}(\widehat{T}) &= (\sigma_2^2+\sigma_3^2)\beta + \sigma_3^2\alpha - \sigma_3^2\end{aligned}$$

From the stationarity conditions

$$\frac{\partial}{\partial\alpha}\text{var}(\widehat{T}) = 0; \quad \frac{\partial}{\partial\beta}\text{var}(\widehat{T}) = 0$$

it follows that

$$\begin{cases} (\sigma_1^2+\sigma_3^2)\alpha + \sigma_3^2\beta - \sigma_3^2 = 0 \\ \\ (\sigma_2^2+\sigma_3^2)\beta + \sigma_3^2\alpha - \sigma_3^2 = 0 \end{cases} \Longrightarrow \begin{cases} \alpha^\circ = \dfrac{\frac{1}{\sigma_1^2}}{\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}+\frac{1}{\sigma_3^2}} \\ \\ \beta^\circ = \dfrac{\frac{1}{\sigma_2^2}}{\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}+\frac{1}{\sigma_3^2}} \end{cases}$$

and thus

$$\gamma^\circ = \frac{\frac{1}{\sigma_3^2}}{\frac{1}{\sigma_1^2}+\frac{1}{\sigma_2^2}+\frac{1}{\sigma_3^2}}$$

$$\delta^\circ = -b_1\alpha^\circ - b_2\beta^\circ - b_3\gamma^\circ$$

Hence the best linear unbiased estimator is given by:

$$\widehat{T}^\circ = \alpha^\circ(x_1-b_1) + \beta^\circ(x_2-b_2) + \gamma^\circ(x_3-b_3)$$

[4 Marks]

d) i) From the answer to Question a), we have $\mathbb{E}[\widehat{T}^\circ] = T$. Then, the average of N such independent unbiased estimators is also unbiased. More formally:

$$\begin{aligned}\mathbb{E}[\widehat{T}_N] &= \mathbb{E}\left[\frac{1}{N}\sum_{i=1}^{N}\widehat{T}^\circ(x_1(i),x_2(i),x_3(i))\right] \\ &= \frac{1}{N}\sum_{i=1}^{N}\mathbb{E}[\widehat{T}^\circ] = \frac{1}{N}\cdot NT = T\end{aligned}$$

Hence $\widehat{T}_N$ still is an unbiased estimator of T. [3 Marks]

ii) Since each triplet $\{x_1(i),x_2(i),x_3(i)\}$ is uncorrelated with each other as was shown before, then $\widehat{T}^\circ(x_1(i),x_2(i),x_3(i))$ values across different i are also uncorelated and identically distributed. Thus:

$$\text{var}[\widehat{\text{T}}_\text{N}] = \text{var}\left[\frac{1}{\text{N}}\sum_{\text{i=1}}^{\text{N}}\widehat{\text{T}}^\circ(\text{i})\right] = \frac{1}{\text{N}^2}\sum_{\text{i=1}}^{\text{N}}\text{var}[\widehat{\text{T}}^\circ(\text{i})]$$

Answers

$$= \frac{1}{N^2} \cdot N\text{var}[\widehat{\mathrm{T}}^{\circ}] = \frac{1}{\mathrm{N}}\text{var}[\widehat{\mathrm{T}}^{\circ}]$$
$$= \frac{1}{N} \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2}}$$

Hence, the variance $\text{var}[\widehat{T}_N]$ shrinks inversely with respect to N, i.e., it converges to 0 as $N \to \infty$.

[3 Marks]

iii) To compute the minimum sample size $N_{\min}$ such that

$$\text{var}[\widehat{T}_{N_{\min}}] < 0.02$$

we solve:

$$\frac{1}{N} \frac{1}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2}} < 0.02$$

Hence the minimum sample size N is the closest integer larger than

$$\frac{50}{\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2}}.$$

[3 Marks]

Answers